

Question 1

P1

A soccer team consists of 14 players. Determine the number of ways a captain and an assistant captain can be selected.

Solution

$${}_{14}P_2 = 182$$

or

$$\underline{14} \cdot \underline{13} = 182$$

1 mark

Question 2

R10

The pH of a person's blood can be found using the formula

$$\text{pH} = 6.1 + \log\left(\frac{B}{C}\right)$$

where B represents the concentration of bicarbonate in the blood, in mEq/L

C represents the concentration of carbonic acid in the blood, in mEq/L

Kansas has a blood pH of 7.41. In her blood sample, the concentration of carbonic acid is 1.41 mEq/L. Determine the concentration of bicarbonate in her blood.

Express the answer correct to one decimal place.

Solution

Method 1

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$10^{1.31} = \frac{B}{1.41} \quad 1 \text{ mark for exponential form}$$

$$B = 28.788 \ 505\dots$$

$$B = 28.8 \text{ mEq/L} \quad \frac{1}{2} \text{ mark for solving for } B$$

2 marks

Method 2

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$1.31 = \log B - \log 1.41 \quad 1 \text{ mark for quotient law}$$

$$1.31 + \log 1.41 = \log B$$

$$1.4592\dots = \log B$$

$$B = 28.788 \ 505\dots$$

$$B = 28.8 \text{ mEq/L} \quad \frac{1}{2} \text{ mark for solving for } B$$

2 marks

Solve $(2 \tan x - 1)(\tan x + 1) = 0$ where $x \in \mathbb{R}$.

Solution

Method 1

$$2 \tan x - 1 = 0$$

$$\tan x + 1 = 0$$

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$x_r = 0.463\ 647\dots$$

$$x = 0.464$$

$$x = \frac{3\pi}{4}$$

$$x = 3.605$$

$$x = \frac{7\pi}{4}$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value)

$$\left. \begin{array}{l} x = 0.464 + 2k\pi \\ x = 3.605 + 2k\pi \\ x = \frac{3\pi}{4} + 2k\pi \\ x = \frac{7\pi}{4} + 2k\pi \end{array} \right\} k \in \mathbb{Z}$$

1 mark for general solution

3 marks

Method 2

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$x_r = 0.463\ 647\dots$$

$$x = \frac{3\pi}{4}$$

2 marks for solving for x (1 mark for each branch)

$$x = 0.464$$

$$\left. \begin{array}{l} x = 0.464 + k\pi \\ x = \frac{3\pi}{4} + k\pi \end{array} \right\} k \in \mathbb{Z}$$

1 mark for general solution

3 marks

Determine and simplify the 7th term in the binomial expansion of $\left(\frac{2}{x^2} + 3x\right)^9$.

Solution

$$t_7 = {}_9C_6 \left(\frac{2}{x^2}\right)^3 (3x)^6 \quad 2 \text{ marks (1 mark for } {}_9C_6; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 84 \left(\frac{8}{x^6}\right) (729x^6)$$

$$= 489\,888$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

Question 5

T1

A winch, with a diameter of 14 cm, is used to pull a boat out of the water. As the winch rotates, it pulls in a cable. Determine the length of cable that is pulled in if the winch rotates 526° .



Solution

$$\theta = (526) \left(\frac{\pi}{180} \right)$$

1 mark for conversion

$$\theta = \frac{263\pi}{90}$$

$$\theta = 9.180\,431\dots$$

$$s = \theta r$$

$$s = \left(\frac{263\pi}{90} \right) (7)$$

1 mark for substitution

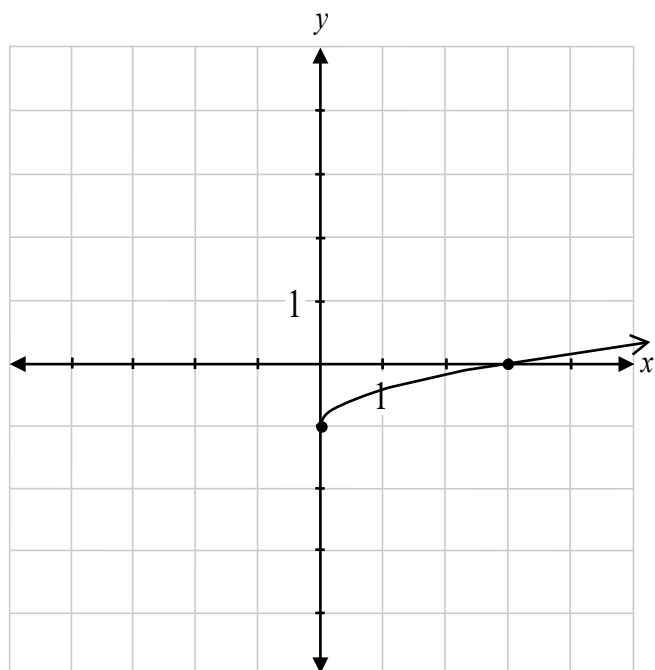
$$s = \frac{1841\pi}{90} \text{ cm}$$

2 marks

or

$$s = 64.263 \text{ cm}$$

Sketch the graph of $f(x) = \sqrt{\frac{1}{3}x} - 1$.

Solution

1 mark for shape of a radical function
1 mark for horizontal stretch
1 mark for vertical translation

3 marks

Question 7

R2, R3, R5

Given $y = f(x)$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection over the y -axis
- vertical stretch by a factor of 3
- horizontal translation 2 units left

Solution

$$g(x) = \underline{3f(-(x+2))}$$

1 mark for horizontal reflection

1 mark for vertical stretch

1 mark for horizontal translation

3 marks

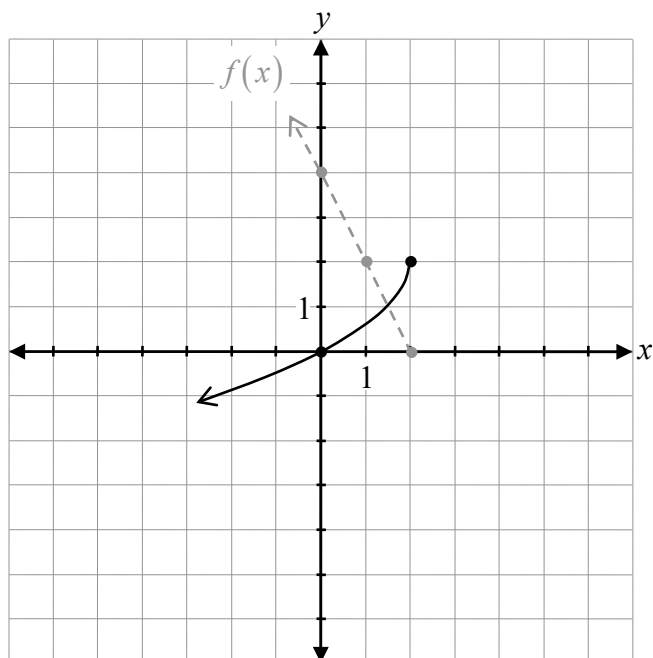
Explain why the graph of $y = \frac{6x+7}{3x+5}$ has a horizontal asymptote at $y = 2$.

Solution

When the degree of the numerator and the denominator are the same, the horizontal asymptote is $y = \frac{a}{b}$ where a is the leading coefficient in the numerator and b is the leading coefficient in the denominator.

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = -\sqrt{f(x)} + 2$.

Solution

1 mark for shape of $\sqrt{f(x)}$

1 mark for vertical reflection

1 mark for vertical translation

3 marks

Using the laws of logarithms, completely expand the given expression.

$$\log\left(\frac{A}{\sqrt[3]{B} \cdot C^4}\right)$$

Solution

$$\log A - \frac{1}{3}\log B - 4\log C$$

1 mark for product law

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

3 marks

Prove the following identity for all permissible values of θ .

$$\cot \theta - \tan \theta = \frac{2 \cos 2\theta}{\sin 2\theta}$$

Solution

Left-Hand Side	Right-Hand Side
$\cot \theta - \tan \theta$ $\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$ $\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}$ $\frac{\cos 2\theta}{\sin \theta \cos \theta}$	$\frac{2 \cos 2\theta}{\sin 2\theta}$ $\frac{2 \cos 2\theta}{2 \sin \theta \cos \theta}$ $\frac{\cos 2\theta}{\sin \theta \cos \theta}$

1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

R11

Express $p(x) = -2x^3 + x^2 + 13x + 6$ in completely factored form.

Solution

$$p(-2) = -2(-2)^3 + (-2)^2 + 13(-2) + 6$$

$$p(-2) = -2(-8) + 4 - 26 + 6$$

$$p(-2) = 0$$

1 mark for identifying one zero of $p(x)$

$\therefore (x + 2)$ is a factor.

$$\begin{array}{r|rrrr} -2 & -2 & 1 & 13 & 6 \\ & \downarrow & 4 & -10 & -6 \\ \hline & -2 & 5 & 3 & 0 \end{array}$$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x + 2)(-2x^2 + 5x + 3)$$

$$p(x) = -(x + 2)(2x^2 - 5x - 3)$$

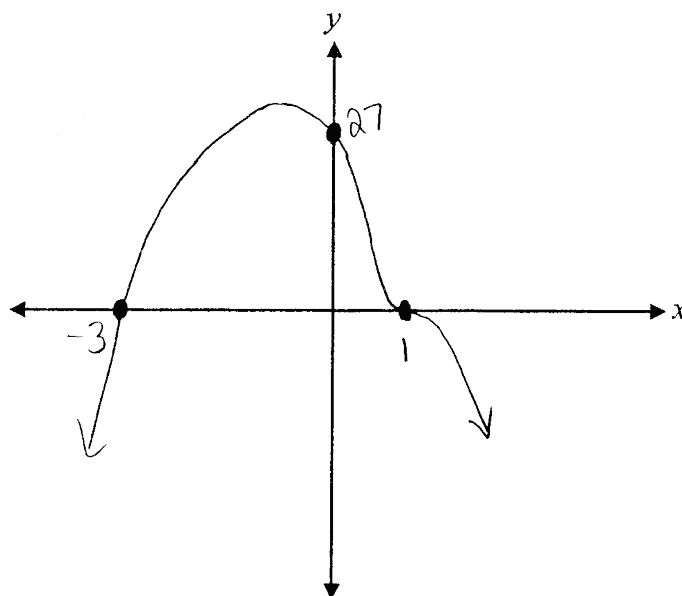
$$p(x) = \underline{-(x + 2)(2x + 1)(x - 3)}$$

1 mark for consistent product of factors

3 marks

Kaitlyn was asked to sketch the graph of the polynomial function $p(x) = -(x-1)(x+3)^3$.

Kaitlyn's graph:



Describe an error on her graph.

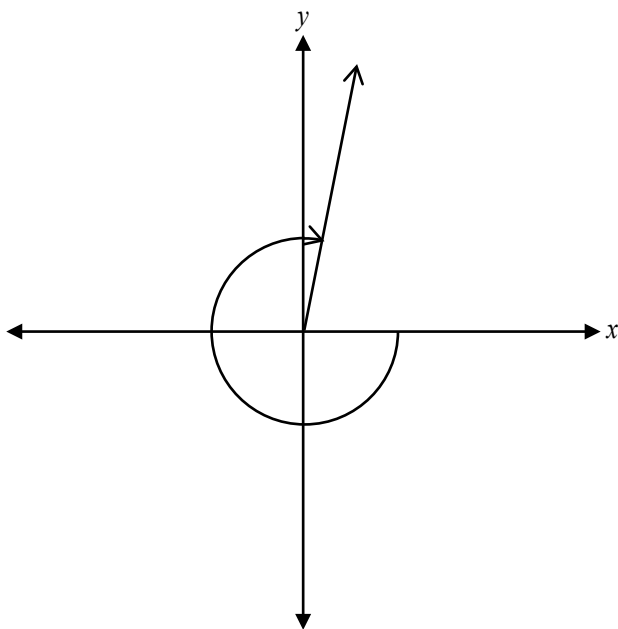
Solution

Kaitlyn's graph shows a multiplicity of 3 at $x = 1$ instead of at $x = -3$.

1 mark

Sketch the angle of -5 radians in standard position.

Solution



$\frac{1}{2}$ mark for an appropriate angle in quadrant I

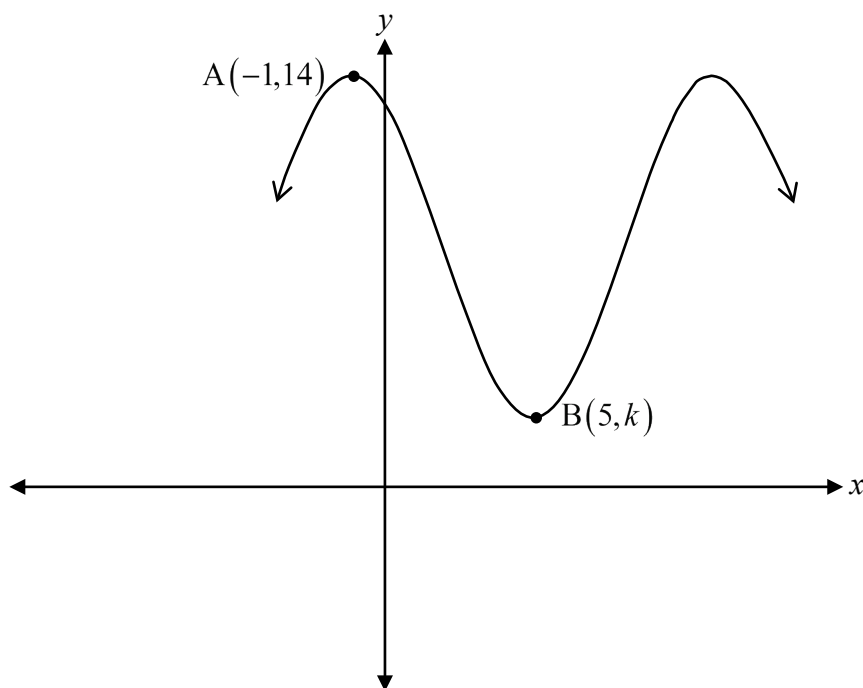
$\frac{1}{2}$ mark for the correct direction of an angle in standard position

1 mark

Question 15

T4

The graph of a sinusoidal function is sketched below. The point A is a maximum and the point B is a minimum. State the value of k if the amplitude of the function is 6.

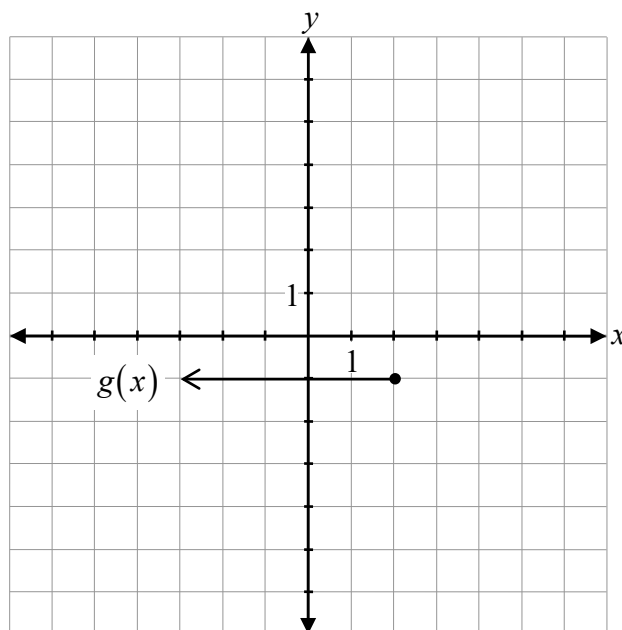
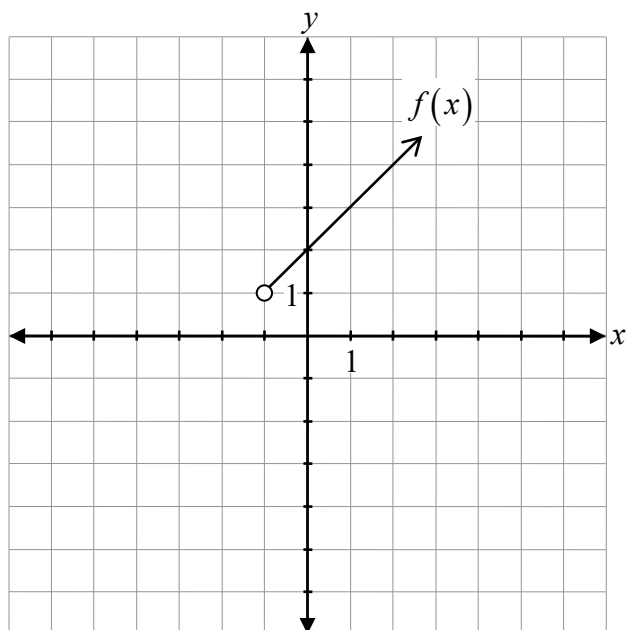


Solution

$$k = 2$$

1 mark

Given the graphs of $f(x)$ and $g(x)$, state the domain of $(f + g)(x)$.

**Solution**

Domain: $\{x \mid -1 < x \leq 2, x \in \mathbb{R}\}$

1 mark

or

Domain: $(-1, 2]$

Question 17

R2, R6

Given that the point $(-13, 7)$ is on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = f^{-1}(x+2)$.

a) $(7, -11)$

b) $(5, -13)$

c) $(7, -15)$

d) $(9, -13)$

Question 18

R12

Given that $x = 0$, $x = 1$, $x = 2$ and $x = -3$ are zeros of the polynomial function $p(x)$, identify a possible equation for $p(x)$.

a) $p(x) = -3x(x-1)(x-2)$

b) $p(x) = (x+1)(x+2)(x-3)^2$

c) $p(x) = x(x+3)^2(x-1)(x-2)$

d) $p(x) = (x-2)(x-1)^3(x+3)$

Question 19

T3

Identify the interval where the angle, θ , could exist if $\cos \theta = -\frac{4}{5}$ and $\csc \theta = \frac{5}{3}$.

a) $\left(0, \frac{\pi}{2}\right)$

b) $\left(\frac{\pi}{2}, \pi\right)$

c) $\left(\pi, \frac{3\pi}{2}\right)$

d) $\left(\frac{3\pi}{2}, 2\pi\right)$

Question 20

T6

Identify a non-permissible value of θ in the identity, $\frac{\cot \theta}{\csc \theta} = \cos \theta$.

a) π

b) $\frac{\pi}{4}$

c) $\frac{3\pi}{2}$

d) $\frac{\pi}{2}$

Question 21

R10

Identify the solution of the equation, $e^{2x} = 16$.

a) $x = \ln 8$

b) $x = 8$

c) $x = \frac{\ln 16}{2}$

d) $x = \frac{\log 16}{2}$

Question 22

P3

A committee of 6 people is to be selected from 90 students. Identify an expression that represents the number of possible committees if two of the students, Evelyn and Wayne, must be on the committee.

a) ${}_{90}C_4$

b) ${}_{90}C_6$

c) ${}_{88}C_6$

d) ${}_{88}C_4$

Question 23

P2

Identify the expression for the number of arrangements of the letters in the word RUSSELL.

a) $\frac{7!}{4!}$

b) $7!$

c) $\frac{7!}{2!}$

d) $\frac{7!}{2!2!}$

Question 24

P4

Given the row of Pascal's triangle below, identify the third value of the next row.

1 7 21 35 35 21 7 1

a) 8

b) 28

c) 56

d) 70

Question 25

R9

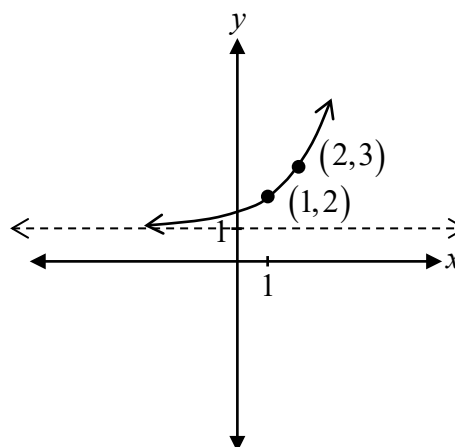
Identify the equation represented by the graph sketched below.

a) $y = \left(\frac{1}{2}\right)^x + 1$

b) $y = 2^{x-1} + 1$

c) $y = 2^{x+1} + 1$

d) $y = \left(\frac{1}{2}\right)^{x+1} + 1$



Match the following functions with their correct graphs.

Place the appropriate letter in this column.

Solution

$$f(x) = \frac{1}{x-2}$$

C

$$g(x) = \frac{-x}{x-2}$$

D

$$h(x) = \frac{2x}{x-1}$$

A

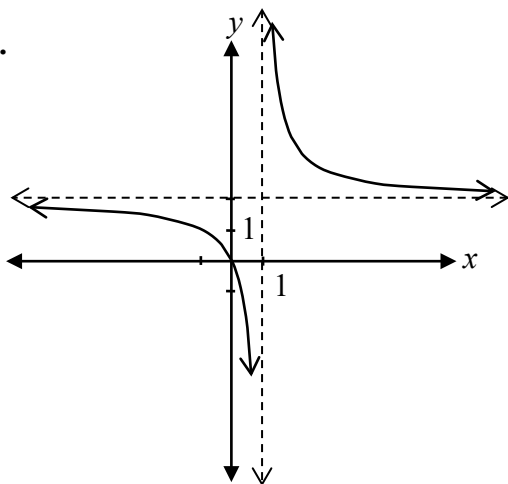
$$k(x) = \frac{2x}{x+1}$$

B

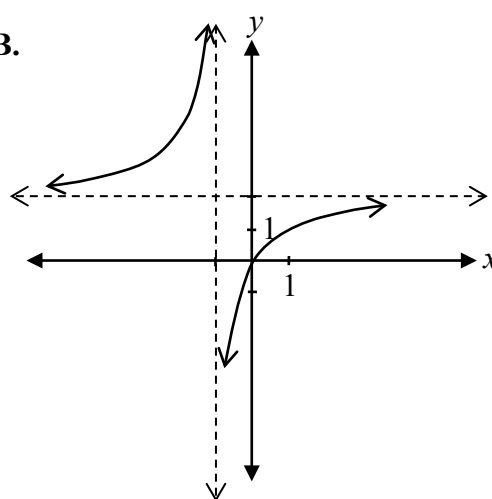
½ mark for each answer

2 marks

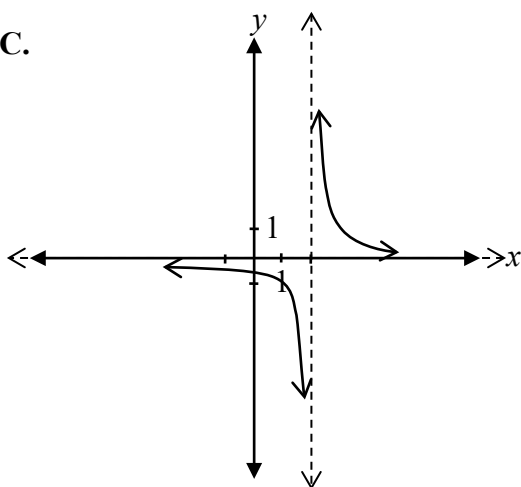
A.



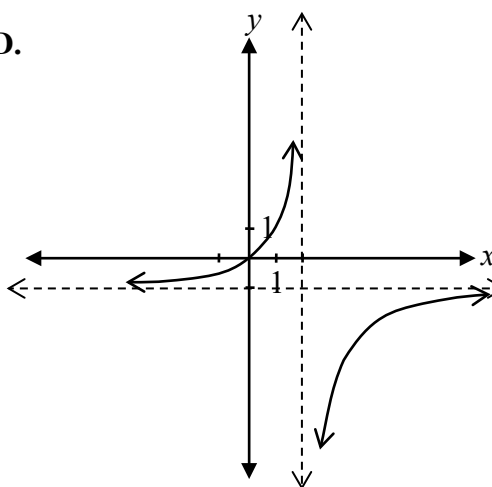
B.



C.



D.



Evaluate.

$$\sin^2\left(\frac{11\pi}{3}\right)\cot\left(\frac{\pi}{4}\right) + \csc\left(\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{-\sqrt{3}}{2}\right)^2 (1) + (-2)$$

1 mark for $\sin\left(\frac{11\pi}{3}\right)$ (½ mark for quadrant; ½ mark for value)

$$\left(\frac{3}{4}\right)(1) - 2$$

1 mark for $\cot\left(\frac{\pi}{4}\right)$ (½ mark for quadrant; ½ mark for value)

$$\frac{3}{4} - 2$$

1 mark for $\csc\left(\frac{11\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

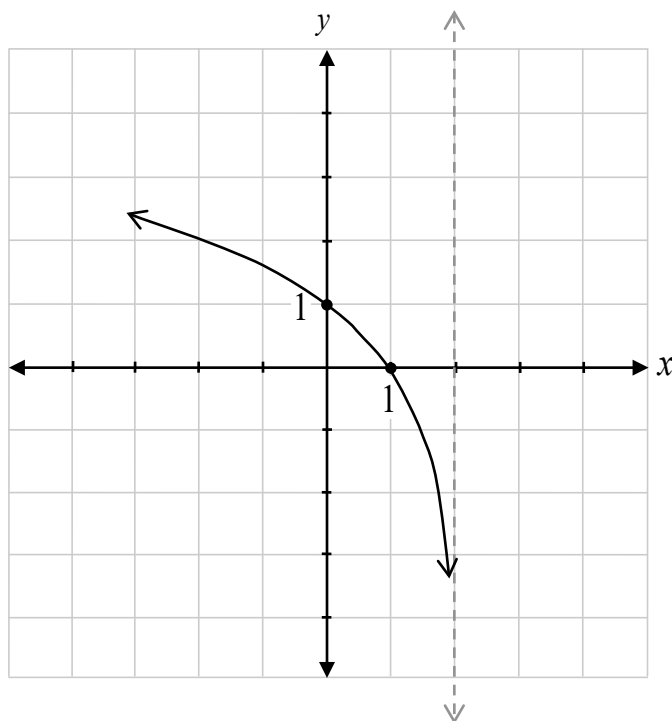
$$-\frac{5}{4}$$

3 marks

or

$$-1\frac{1}{4}$$

Sketch the graph of $y = \log_2[-(x-2)]$.

Solution

1 mark for shape of a logarithmic function

1 mark for horizontal reflection

1 mark for horizontal translation

3 marks

Justify that ${}_{12}C_7 = {}_{12}C_5$.

Solution

$$\begin{aligned} {}_{12}C_7 &= \frac{12!}{(12-7)!7!} & {}_{12}C_5 &= \frac{12!}{(12-5)!5!} \\ &= \frac{12!}{5!7!} & &= \frac{12!}{7!5!} \end{aligned}$$

$$\frac{12!}{5!7!} = \frac{12!}{7!5!}$$

1 mark

Solve, algebraically.

$$\left(\frac{1}{4}\right)^{3x} = 8^{x-3}$$

Solution

$$\left(2^{-2}\right)^{3x} = \left(2^3\right)^{x-3}$$

1 mark for changing to a common base ($\frac{1}{2}$ mark for each side)

$$2^{-6x} = 2^{3x-9}$$

$\frac{1}{2}$ mark for exponent law

$$-6x = 3x - 9$$

$\frac{1}{2}$ mark for equating exponents

$$-9x = -9$$

$$x = 1$$

2 marks

State the domain and range of the function $y = -\sqrt{x-5} + 4$.

Solution

Domain: $\{x|x \geq 5, x \in \mathbb{R}\}$ 1 mark for domain

Range: $\{y|y \leq 4, y \in \mathbb{R}\}$ 1 mark for range

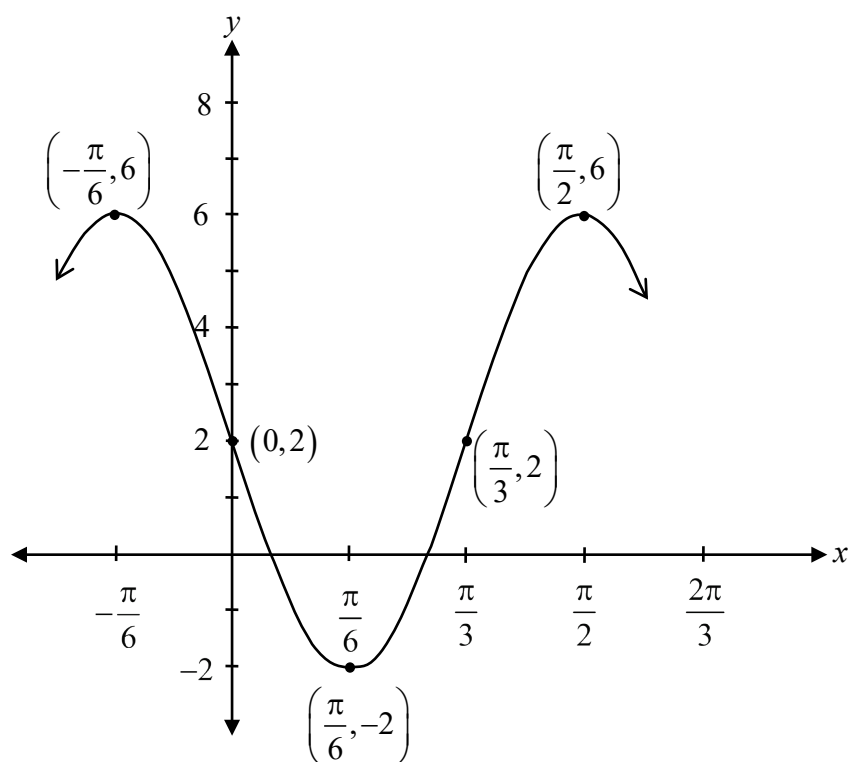
or

Domain: $[5, \infty)$ 1 mark for domain

Range: $(-\infty, 4]$ 1 mark for range

2 marks

Sketch at least one period of the graph of $y = 4\cos\left[3\left(x + \frac{\pi}{6}\right)\right] + 2$.

Solution

1 mark for shape of a sinusoidal function with correct amplitude

1 mark for period

1 mark for horizontal translation

1 mark for vertical translation

4 marks

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has zeros at -5 and 1
- all zeros of $p(x)$ have a multiplicity of 2
- the graph of $p(x)$ has a y -intercept at 75

Solution

$$p(x) = a(x+5)^2(x-1)^2$$

1 mark for multiplicity of 2 at $x = -5$ and $x = 1$

$$75 = a(5)^2(-1)^2$$

$$75 = 25a$$

$$a = 3$$

1 mark for consistent value of a

$$p(x) = \underline{3(x+5)^2(x-1)^2}$$

2 marks

Vitaly was asked to solve the exponential equation $5^{2x+7} = 3 \cdot 7^x$.

Vitaly's work:

$$\begin{aligned}\log(5^{2x+7}) &= \log(3 \cdot 7^x) \\ (2x+7)\log 5 &= \log(21^x) \\ 2x\log 5 + 7\log 5 &= x\log 21 \\ 7\log 5 &= x\log 21 - 2x\log 5 \\ 7\log 5 &= x(\log 21 - 2\log 5) \\ x &= \frac{7\log 5}{\log 21 - 2\log 5}\end{aligned}$$

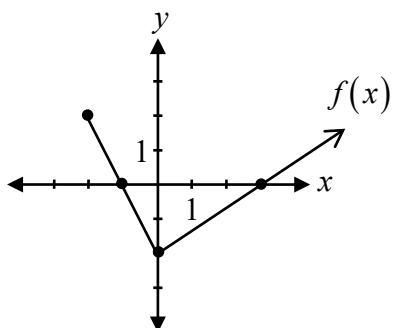
Describe his error.

Solution

In step 2, Vitaly incorrectly simplified $3 \cdot 7^x$ to 21^x .
He should have applied the product law of logarithms.

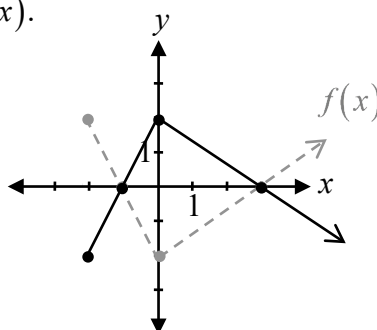
1 mark

Given the graph of $f(x)$,



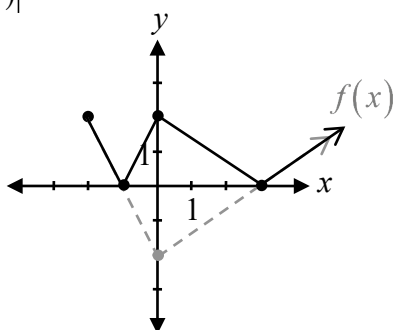
Solution

a) sketch the graph of $g(x) = -f(x)$.



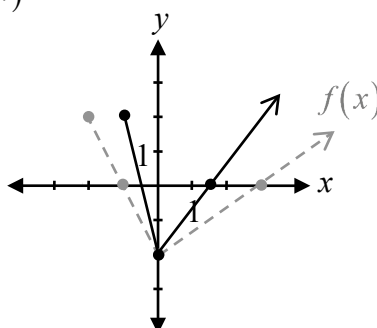
1 mark

b) sketch the graph of $k(x) = |f(x)|$.



1 mark

c) sketch the graph of $r(x) = f(2x)$.



1 mark

Determine the exact value of $\sin\left(\frac{17\pi}{12}\right)$.

Solution

$$\sin\left(\frac{17\pi}{12}\right) = \sin\left(\frac{\pi}{6} + \frac{5\pi}{4}\right)$$

$$= \sin\left(\frac{\pi}{6}\right)\cos\left(\frac{5\pi}{4}\right) + \cos\left(\frac{\pi}{6}\right)\sin\left(\frac{5\pi}{4}\right)$$

1 mark for substitution into correct identity

$$= \left(\frac{1}{2}\right)\left(\frac{-\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{-\sqrt{2}}{2}\right)$$

2 marks for exact values ($\frac{1}{2}$ mark for each)

$$= \frac{-\sqrt{2} - \sqrt{6}}{4}$$

3 marks

Note:

- Other combinations are possible.

Question 37

R1

Given $f(x) = \sqrt{x+3}$ and $g(x) = x-8$,

a) determine the value of $g(f(1))$.

b) justify that the value of $(f-g)(-5)$ is undefined.

Solution

a) $f(1) = 2$ ½ mark for the value of $f(1)$

$$g(2) = -6$$

$g(f(1)) = -6$ ½ mark for consistent value of $g(f(1))$

1 mark

b) The domain of $f(x)$ is $x \geq -3$; $x = -5$ is outside the domain.

or

$$\begin{aligned} f(-5) &= \sqrt{-5+3} \\ &= \sqrt{-2} \end{aligned}$$

It is not possible to evaluate the square root of a negative radicand, therefore $f(-5)$ is undefined.

1 mark

Solve, algebraically.

$$48 \binom{n}{4} = {}_{(n+1)}P_4$$

Solution

$$48 \left(\frac{n!}{(n-4)!4!} \right) = \frac{(n+1)!}{((n+1)-4)!}$$

1 mark for substitution into equation (½ mark for each side)

$$48 \left(\frac{n!}{(n-4)!4!} \right) = \frac{(n+1)!}{(n-3)!}$$

$$\frac{48}{4!} = \frac{(n-4)!(n+1)!}{(n-3)!n!}$$

$$\frac{48}{24} = \frac{\cancel{(n-4)!} (n+1) \cancel{(n)!}}{(n-3) \cancel{(n-4)!} \cancel{(n)!}}$$

1 mark for factorial expansion
½ mark for simplification of factorials

$$2 = \frac{n+1}{n-3}$$

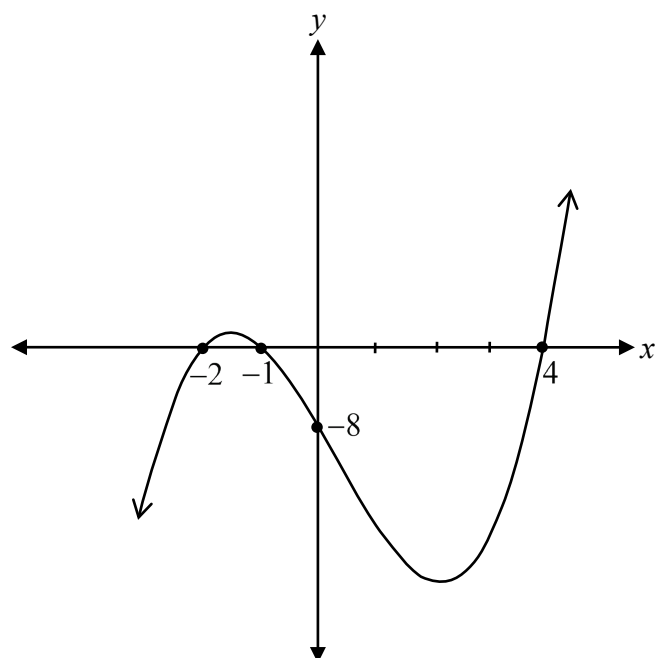
$$2n - 6 = n + 1$$

$$n = 7$$

½ mark for solving for n

3 marks

Sketch the graph of $p(x) = (x+1)(x+2)(x-4)$.

Solution

1 mark for x -intercepts
 $\frac{1}{2}$ mark for y -intercept
 $\frac{1}{2}$ mark for end behaviour

2 marks

Evaluate.

$$\log_5(\log_2 32)$$

Solution

$$\log_5 5$$

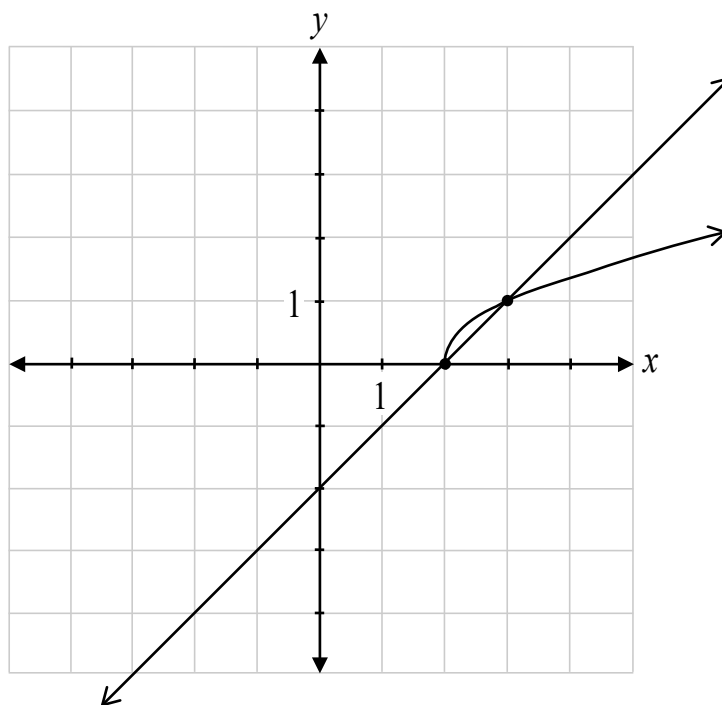
$\frac{1}{2}$ mark for evaluating $\log_2 32$

$$1$$

$\frac{1}{2}$ mark for consistent value

1 mark

Describe how to use the graph to solve the equation $\sqrt{x-2} = x-2$.

**Solution**

State the x -coordinates of the points of intersection.

1 mark

Verify, by substitution, that the equation $\frac{\sec \theta}{\tan \theta + \cot \theta} = \sin 2\theta$ is true for $\theta = \frac{\pi}{3}$.

Solution

Left-Hand Side	Right-Hand Side
$\frac{\sec \frac{\pi}{3}}{\tan \frac{\pi}{3} + \cot \frac{\pi}{3}}$ $\frac{2}{\sqrt{3} + \frac{1}{\sqrt{3}}}$ $\frac{2}{\frac{3+1}{\sqrt{3}}}$ $\frac{2\sqrt{3}}{4}$ $\frac{\sqrt{3}}{2}$	$\sin \frac{2\pi}{3}$ $\frac{\sqrt{3}}{2}$
	½ mark for substitution 2 marks for exact values (½ mark for each) ½ mark for simplification
LHS = RHS	

3 marks

Describe the transformation(s) to the graph of $y = 3^x$ which results in a graph with a y -intercept of -2 .

Solution

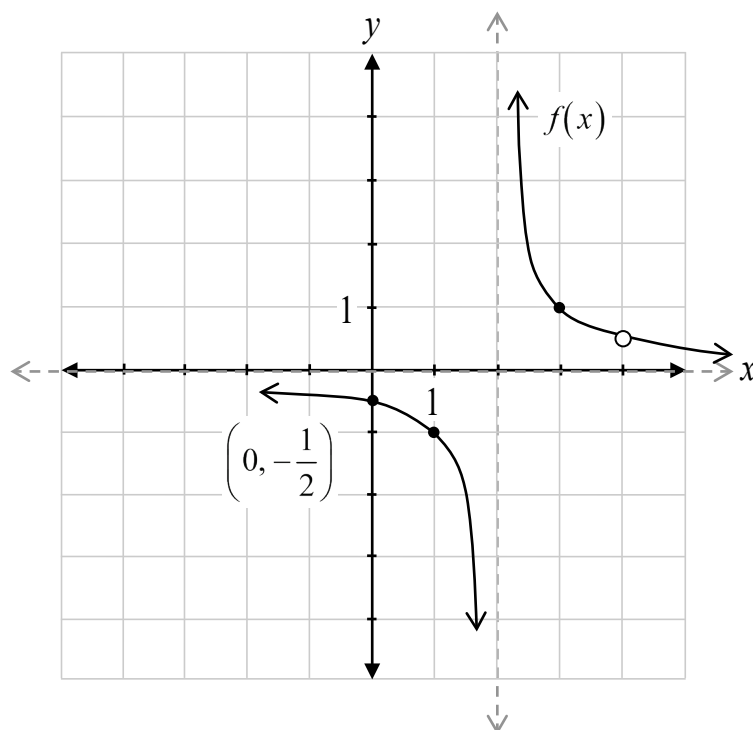
A vertical translation down 3 units.

1 mark

Note:

- Other answers are possible.

Determine the equation of the rational function represented by the graph.



Solution

$$f(x) = \frac{x-4}{(x-2)(x-4)}$$

1 mark for factor of $(x-2)$ in denominator

1 mark for factor of $(x-4)$ in both numerator and denominator

2 marks

Justify that $f(x) = 3x + 2$ and $g(x) = \frac{x-3}{2}$ are not inverses of each other.

Solution**Method 1**

$$\text{Let } y = f(x)$$

$$y = 3x + 2$$

$$\text{Let } y = f^{-1}(x)$$

$$x = 3y + 2$$

1 mark for switching x and y

$$\frac{x-2}{3} = y$$

$$\frac{x-2}{3} = f^{-1}(x)$$

$$g(x) \neq f^{-1}(x)$$

1 mark for justification

2 marks**Method 2**

If they are inverses, then $g(f(x)) = x$, and $f(g(x)) = x$.

$$\begin{aligned} g(3x+2) &= \frac{3x+2-3}{2} \\ &= \frac{3x-1}{2} \end{aligned}$$

1 mark for $g(f(x))$ or $f(g(x))$

Since $g(f(x)) \neq x$, they are not inverses of each other.

1 mark for justification

2 marks

The terminal arm of an angle, θ , in standard position passes through the point $(-3, 6)$ and intersects the unit circle at the point $P(\theta)$. Determine the coordinates of the point $P(\theta)$.

Solution

$$x^2 + y^2 = r^2$$

$$(-3)^2 + (6)^2 = r^2 \quad \frac{1}{2} \text{ mark for substitution}$$

$$9 + 36 = r^2$$

$$45 = r^2$$

$$\sqrt{45} = r \quad \frac{1}{2} \text{ mark for solving for } r$$

$$P(\theta) = \left(\frac{-3}{\sqrt{45}}, \frac{6}{\sqrt{45}} \right) \quad \begin{array}{l} 1 \text{ mark for the coordinates of } P(\theta) \\ (\frac{1}{2} \text{ mark for } x\text{-coordinate; } \frac{1}{2} \text{ mark for } y\text{-coordinate}) \end{array}$$

or

$$P(\theta) = \left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$$

or

$$P(\theta) = \left(\frac{-\sqrt{5}}{5}, \frac{2\sqrt{5}}{5} \right)$$

2 marks

Determine the exact value of $2\sin\theta\cos\theta$ if $\theta = \frac{\pi}{12}$.

Solution

$2\sin\theta\cos\theta = \sin 2\theta$ 1 mark for substitution into correct identity

$$= \sin\left(2\left(\frac{\pi}{12}\right)\right)$$

$$= \sin\left(\frac{\pi}{6}\right)$$

$$= \frac{1}{2}$$

1 mark for consistent value

2 marks